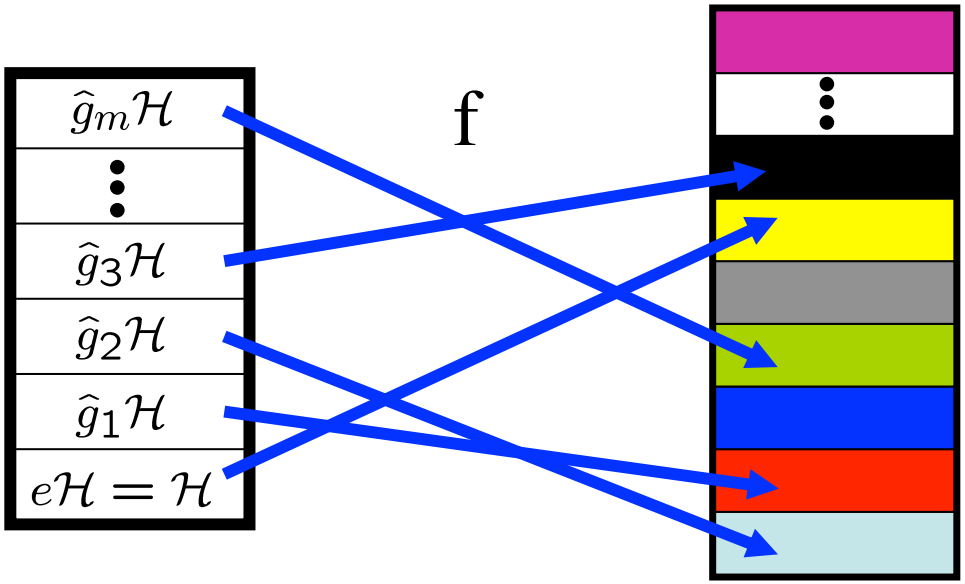
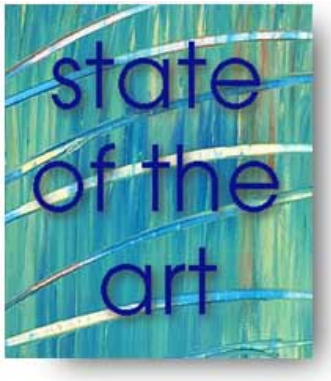
The Hidden Subgroup Conjugacy Problem and Beyond

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State of the Art



Quantum Algorithms

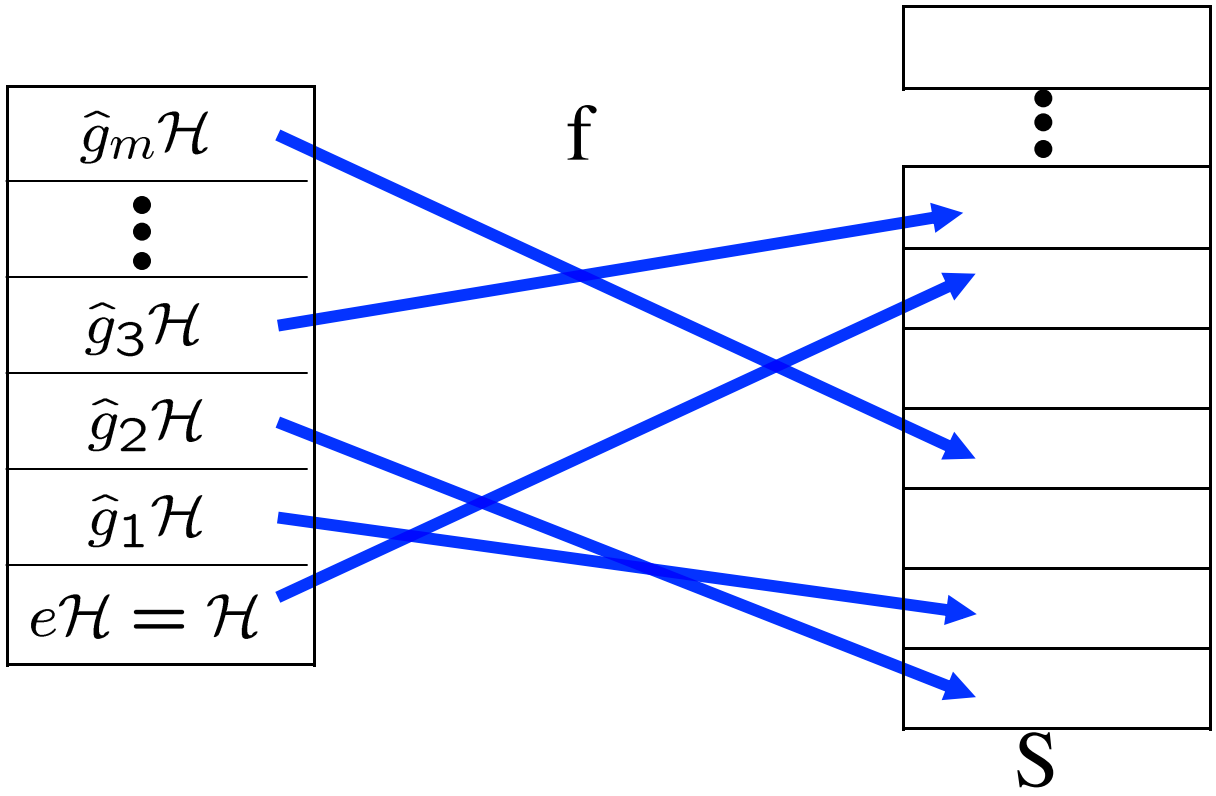
- Factoring, discrete log [Shor 94]
- Unstructured search [Grover 96]
- Various hidden subgroup problems [Long List]
- Pell's equation [Hallgren 02]
- Hidden shift problems [van Dam, Hallgren, Ip 03 Childs, van Dam 05]
- Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- …

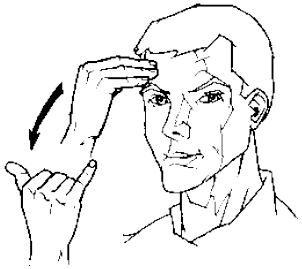
The Hidden Subgroup Problem

Setup: Fix a group $\mathcal{G}$ (known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ (unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$. We say that f hides $\mathcal{H}$.

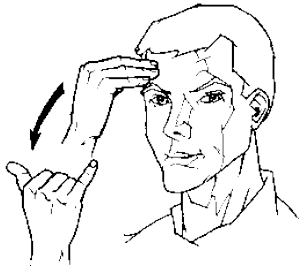
Problem: Find a generating set for $\mathcal{H}$ ("identify $\mathcal{H}$").

Efficient: If runs in time polylog($|\mathcal{G}|$)





Why the Fuss?



Solved Problems (Shor 1994)

efficient algorithm for HSP over $\mathbb{Z}_N$ → efficient algorithm for factoring whole numbers

efficient algorithm for HSP over $\mathbb{Z}_N \times \mathbb{Z}_N$ → efficient algorithm for the discrete logarithm problem

Open Problems

efficient algorithm for HSP over $\mathcal{S}_n \wr \mathbb{Z}_2$ (or $\mathcal{S}_{2n}$) → efficient algorithm for the graph isomorphism problem

efficient "standard" quantum algorithm for HSP over dihedral group $\mathcal{D}_N$ → efficient algorithm for the poly(*n*) unique shortest vector in a lattice problem

Hidden Subgroup Successes

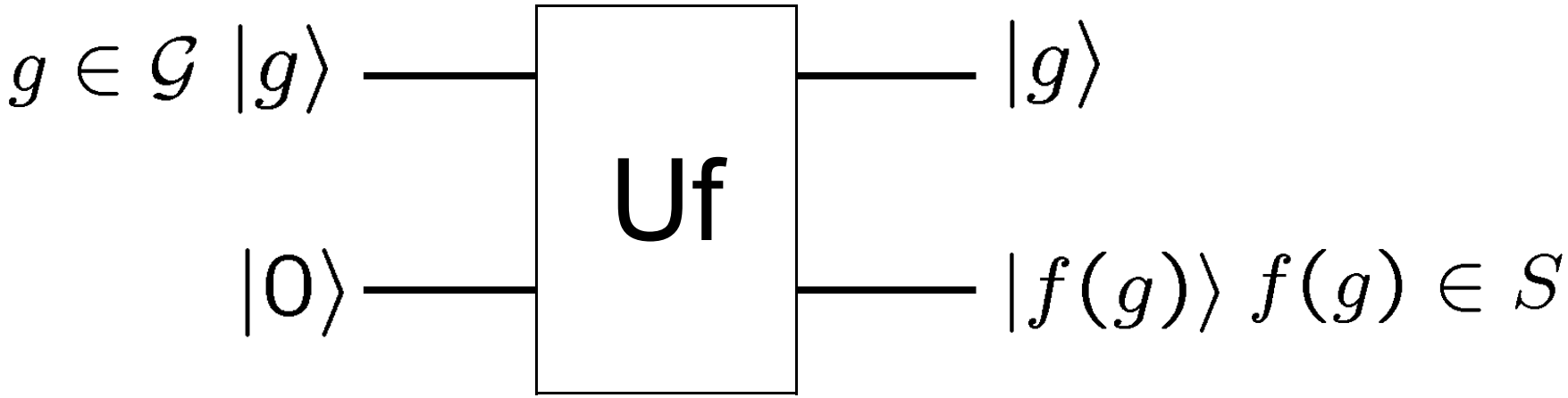
- Abelian groups [Shor 94; Kitaev 95]
- $\mathbb{Z}_2^n \rtimes \mathbb{Z}_2$ [Püschel, Rötteler, Beth 98]
- Normal subgroups [Hallgren, Russell, Ta-Shma 00]
- "Amost Abelian" groups [GSVV 01]
- $\mathbb{Z}_{p^k}^n \rtimes \mathbb{Z}_2$ [FIMMS 02]
- "Weak Hamiltonian" groups [Gavinsky 04]
- Some q-hedral groups [MRRS 04]
- [Inui, Le Gall 04]
- Some $\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$ groups
- Heisen [Bacon, Childs, and van Dam 05]

$\mathbb{Z}_p^k \rtimes \mathbb{Z}_p$

Why **THESE** successes?

Standard Query Model

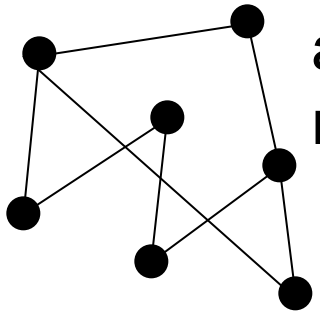
Oracle:



Query in equal superposition over group elements

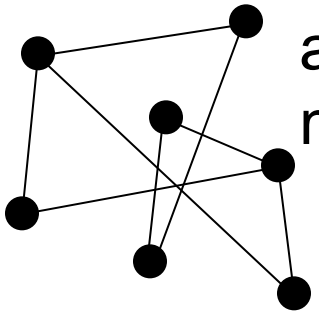
$$\frac{1}{\sqrt{|\mathcal{G}|}} \sum_{g \in \mathcal{G}} |g\rangle|0\rangle \rightarrow \sum_{g \in \mathcal{G}} \frac{1}{\sqrt{|\mathcal{G}|}} |g\rangle|f(g)\rangle$$

Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



adjacency matrix A_2

$G_2 = (V_2, E_2)$

Problem: Is there a 1-to-1 mapping p from the vertices V_1 to the vertices V_2 such that the edges are respected, $p(E_1) = E_2$?

Mapping to the HSP:

$$f : \mathcal{S}_n \wr \mathcal{S}_2 \rightarrow \mathbb{Z}_{n(2n-1)}$$

$$f((\pi_1, \pi_2, \tau)) = \tau(\pi_1(A_1) \oplus \pi_2(A_2))$$

Notice now we are only worried if hidden subgroup has a nontrivial τ

Conjugate Subgroups

Two subgroups, $\mathcal{H}_1$ and $\mathcal{H}_2$ are conjugate if there exists an element of the group $g \in \mathcal{G}$ such that

$$\mathcal{H}_1 = \{ghg^{-1}, h \in \mathcal{H}_2\}$$

This is an equivalence relation among the subgroups of $\mathcal{G}$

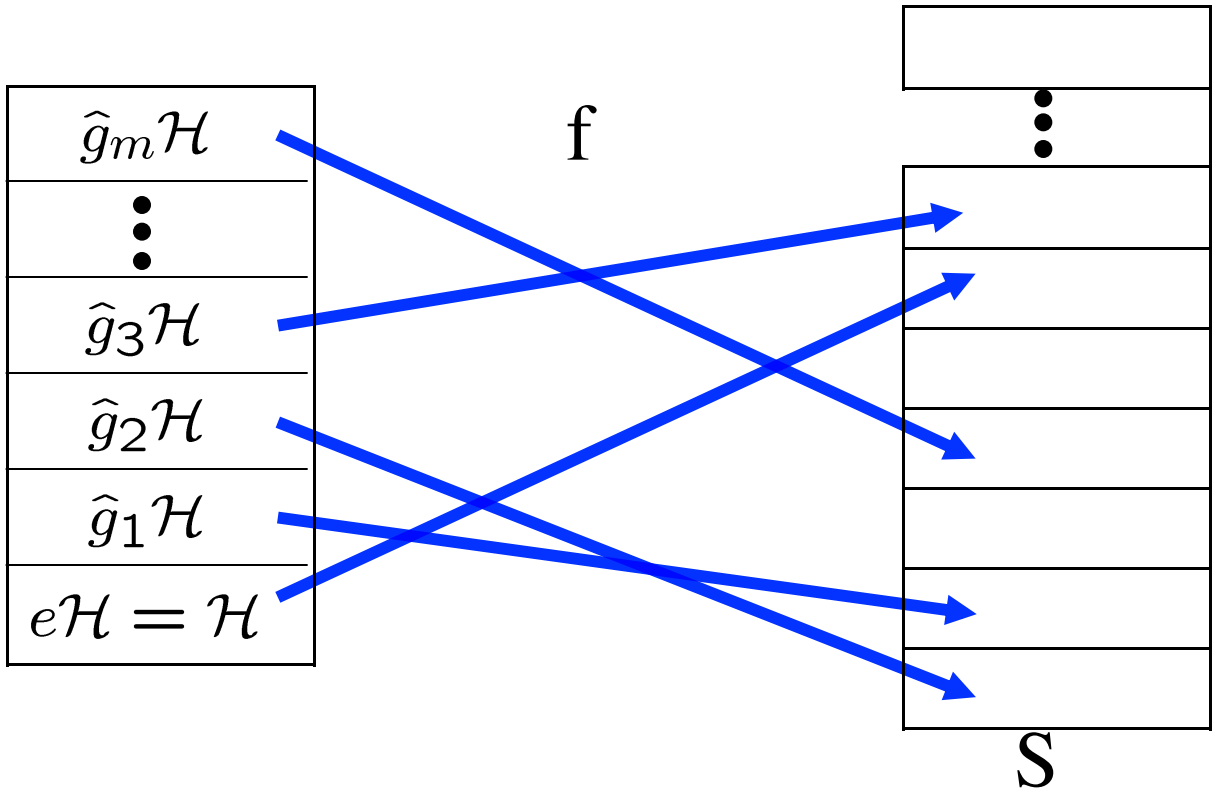
$$\mathcal{H}_i \equiv_{\mathcal{G}} \mathcal{H}_j$$

Hidden Subgroup Conjugacy Problem

Setup: Fix a group $\mathcal{G}$ known) and a subgroup $\mathcal{H} \subset \mathcal{G}$ unknown). Given query access to a function $f : \mathcal{G} \to S$ that is constant and distinct on left (right) cosets of $\mathcal{H}$ in $\mathcal{G}$ We say that f hides $\mathcal{H}$

Problem: identify the conjugate subgroup class of $\mathcal{H}$

Efficient: If runs in time polylog($|\mathcal{G}|$)



Relationships

How do the hidden subgroup problem (HSP) and the hidden subgroup conjugacy problem (HSCP) differ?

1. If group is abelian, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

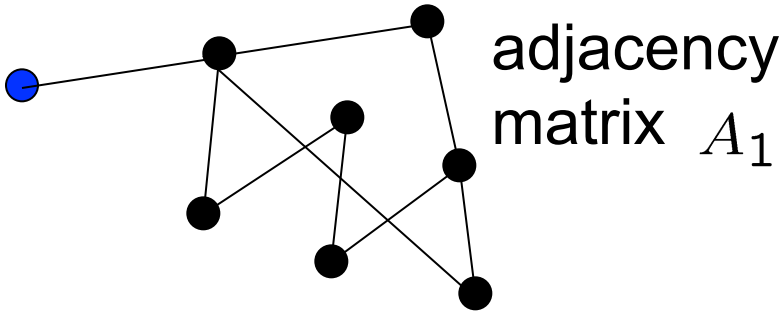
2. If subgroups are normal, HSCP=HSP

All subgroups are only conjugate to themselves

$$g\mathcal{H}g^{-1} = \mathcal{H}$$

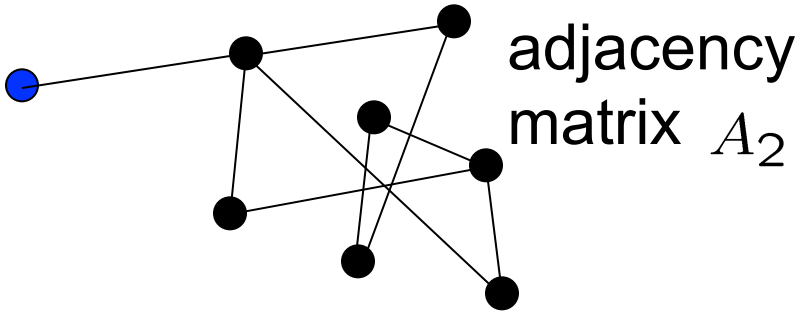
These are two cases where we know quantum computers can efficiently solve the HSP. Coincidence?

Graph Isomorphism



adjacency matrix A_1

$G_1 = (V_1, E_1)$



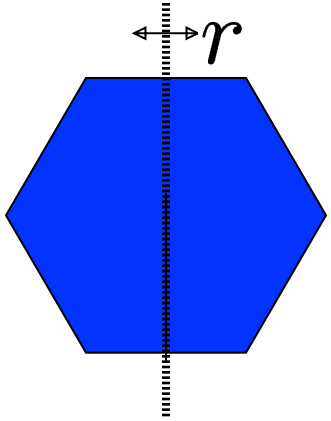
adjacency matrix A_2

$G_2 = (V_2, E_2)$

If we want to find the isomorphism, we can attach extra vertices and edges to detect whether permutation of these vertices is in isomoprhism group.

HSCP $\rightarrow$ solve graph isomorphism and identify the isomorphism

Dihedral Group



s = rotation

r = reflection

Dihedral group $\mathcal{D}_N$

group of symmetries of a *N* sided regular polygon

$$\mathcal{D}_N = \{r^t s^k, t \in \mathbb{Z}_2, k \in \mathbb{Z}_N\}$$

$$r^2 = s^N = rsrs = e$$

Subgroups of $\mathcal{D}_N$

cyclic subgroups:

$$\mathcal{C}_{N/j} = \{e, s^j, s^{2j}, \ldots, s^{-j}\}$$

$$j \in \mathbb{Z}_N \quad j \in \mathbb{Z}_N \quad j|N$$

dihedral subgroups:

$$\mathcal{D}_{N/j,l} = \{e, s^j, \ldots, s^{-j}, rs^l, rs^{j+l}, \ldots, rs^{-j+l}\}$$

$$j \in \mathbb{Z}_N \quad j \in \mathbb{Z}_N \quad j|N \quad l \in \mathbb{Z}_N$$

Dihedral HSP

Theorem [Ettinger, Hoyer 98]

Sufficient to solve Dihedral HSP when subgroups are limited to

$$\mathcal{H}_l = \{e, rs^l\} \quad l \in \mathbb{Z}_N \quad \text{or} \quad \mathcal{T} = \{e\}$$

if N is odd, then the $\mathcal{H}_l = \{e, rs^l\}$ subgroups are all conjugate to each other:

$$s^{-k}\mathcal{H}_l s^k = \{e, rs^{l+2k}\}$$

solving the Dihedral HSCP does not immediately solve the Dihedral HSP…

Modification



Modify the function hiding the subgroup:

$$f : \mathcal{D}_N \to S$$
$$g : \mathcal{D}_N \to \{+1, -1\}$$
$$g(r^t s^k) = (-1)^k$$

left cosets:

$$\mathcal{H}_l = \{e, rs^l\}$$
$$s\mathcal{H}_l = \{s, rs^{l-1}\}$$
$$s^2\mathcal{H}_l = \{s, rs^{l-2}\}$$
$$\vdots$$

If l is even, then (f, g) is constant and distinct on left cosets of $\mathcal{H}_l$

If l is odd, then (f, g) is constant and distinct on left cosets of $\mathcal{T} = \{e\}$

Solving HSCP for new problem thus allows us to distinguish between odd and even subgroups

[Moore 05, personal communication]

Modification

Modify the function hiding the subgroup:

$$f : \mathcal{D}_N \to S$$
$$g_1 : \mathcal{D}_N \to \{+1, -1\}$$
$$g_2 : \mathcal{D}_N \to \{+1, -1\}$$

left cosets:

$$\mathcal{H}_l = \{e, rs^l\}$$
$$s\mathcal{H}_l = \{s, rs^{l-1}\}$$
$$s^2\mathcal{H}_l = \{s, rs^{l-2}\}$$
$$\vdots$$

Using more functions, we can determine 2nd least significant bit of l

Generalizing this allows us to use the Dihedral HSCP to solve the Dihedral HSP (they are polynomial time equivalent.)

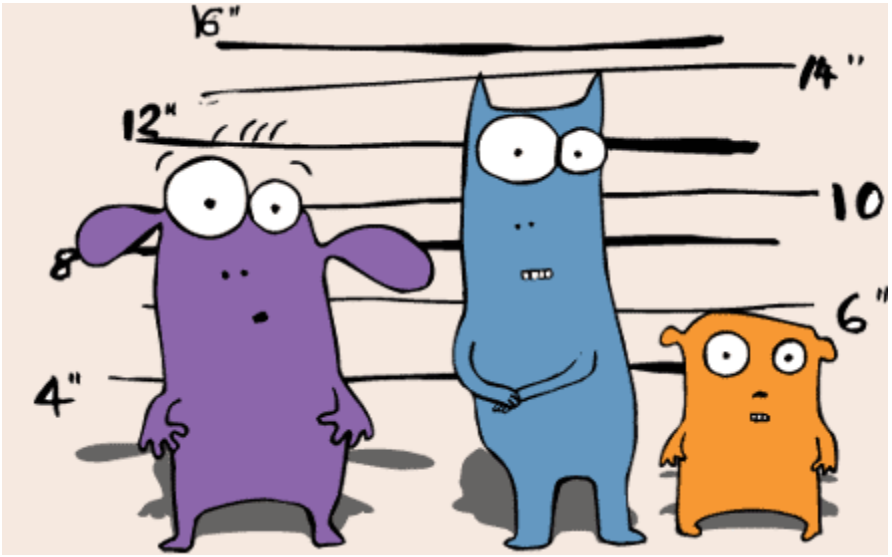
[Moore 05, personal communication]

see also [Kuperberg 03]

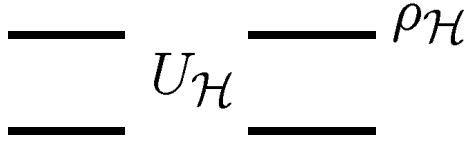
Open question: When are HSP and HSCP equivalent?

Why the HSCP?

The HSCP has more symmetry than the HSP

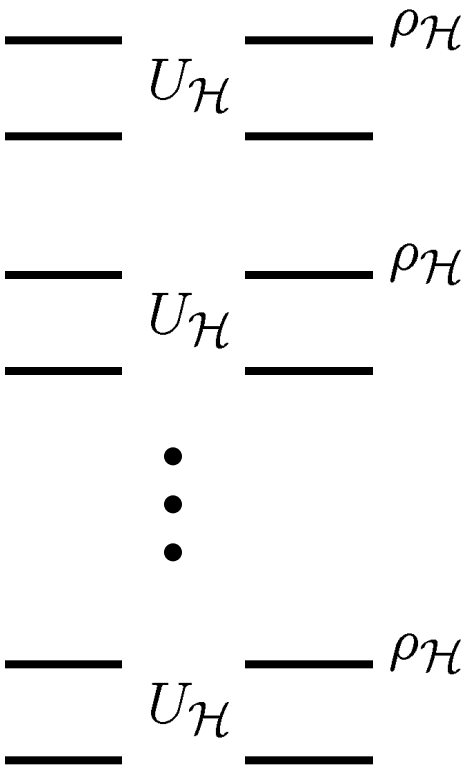


Multiple Queries



If we make a single measurement on the HS state, can we distinguish different $\mathcal{H}$

In general, NO

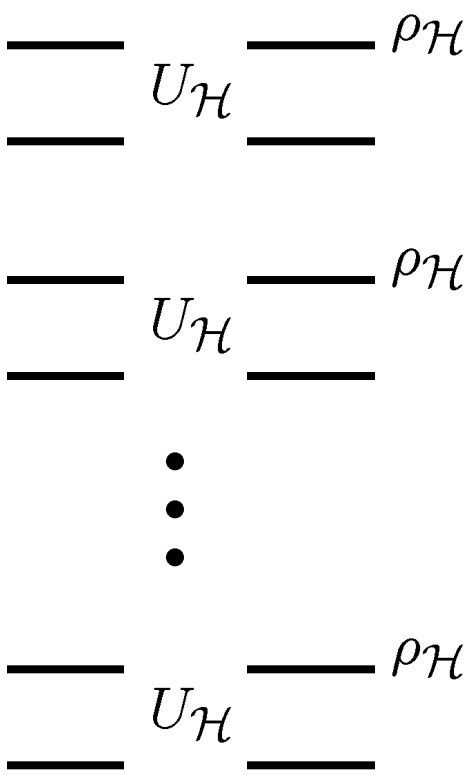


We need multiple queries to distinguish different subgroups

$\rho_{\mathcal{H}}^{\otimes m}$ $m = O(\log |\mathcal{G}|)$ sufficient

[Ettinger, Høyer, Knill 99]

Multiple Queries



$\rho_{\mathcal{H}}^{\otimes m}$

Can we make a measurement on $m < O(\log |\mathcal{G}|)$ copies and learn the subgroup?

In general (and for important case of symmetric group) NO

[Moore, Russell, Hallgren, Roetteler, Sen 05]



Symmetries



left regular representation

$$D_L(g)|g'\rangle = |gg'\rangle$$

right regular representation

$$D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$g, g' \in \mathcal{G}$$

they commute $D_L(g_1)D_R(g_2) = D_R(g_2)D_L(g_1)$

representation theory

$$D_L(g) \cong \bigoplus_{\mu} (D_\mu(g) \otimes I_{d_\mu})$$

$$D_R(g) \cong \bigoplus_{\mu} (I_{d_\mu} \otimes D_\mu(g))$$

sum over all irreps

The transform from the basis of group elements $|g\rangle$ to this block diagonal basis $|\mu, i, j\rangle$ is the Fourier transform over $\mathcal{G}$

$$Q_{\mathcal{G}} = \sum_{\mu} \sum_{i,j=1}^{d_\mu} \sum_{g \in \mathcal{G}} \sqrt{\frac{d_\mu}{|\mathcal{G}|}} [D_\mu(g)]_{i,j} |\mu, i, j\rangle\langle g|$$

Symmetries of HS States

left regular representation

$$D_L(g)|g'\rangle = |gg'\rangle$$

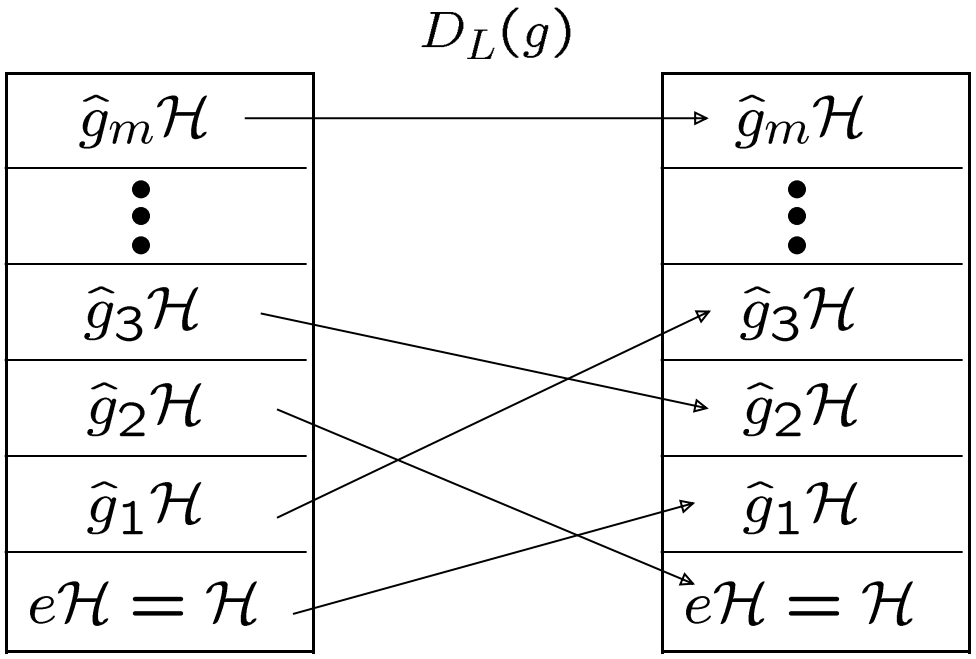
right regular representation

$$D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$g, g' \in \mathcal{G}$$

$$\rho_{\mathcal{H}} = \frac{|\mathcal{H}|}{|\mathcal{G}|} \sum_{g \in \hat{\mathcal{G}}} |g\mathcal{H}\rangle\langle g\mathcal{H}|$$

$$D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$



Symmetries of HS States

left regular representation

$$D_L(g)|g'\rangle = |gg'\rangle$$

right regular representation

$$D_R(g)|g'\rangle = |g'g^{-1}\rangle$$

$$g, g' \in \mathcal{G}$$

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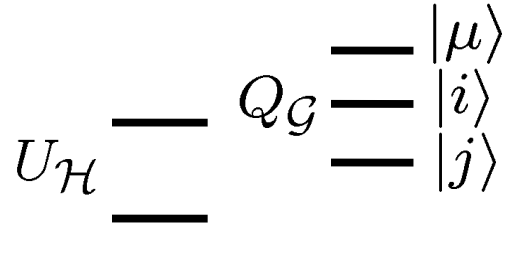
$$D_L(g)\rho_{\mathcal{H}} D_L(g^{-1}) = \rho_{\mathcal{H}}$$

via Schur's lemma this implies

$$\rho_{\mathcal{H}} = \frac{1}{|\mathcal{G}|} \sum_{h \in \mathcal{H}} D_R(h)$$

after applying the Fourier transform

$$\rho_{\mathcal{H}} \cong \bigoplus_{\mu} \left(I_{d_\mu} \otimes \frac{1}{|\mathcal{G}|} \sum_{h \in \mathcal{H}} D_\mu(h) \right)$$



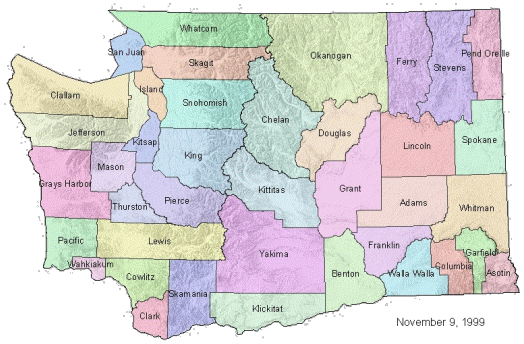
State Distinguishablity

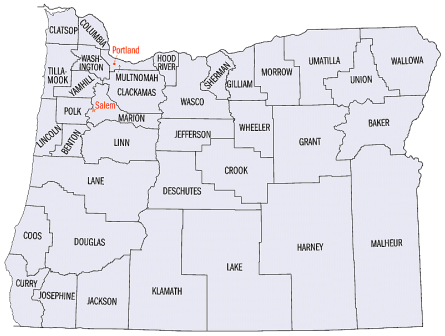
Given $\rho_{\mathcal{H}}$ with probability $p_{\mathcal{H}}$

Find measurement which maximizes probability of successfully identifying $\rho_{\mathcal{H}}$

[Ip 03, Bacon, Childs, van Dam 05, Moore, Russell 05, Bacon, Childs, van Dam 05, Childs, van Dam 05, Childs, Wocjan 05, …]

Often the measurement which maximizing this probability is the “pretty-good-measurement” (PGM)





State Distinguishablity

HSCP: we don't care about which subgroup, just it's conjugacy. New problem

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \rho_{g\mathcal{H}g^{-1}}$$

Probability: $p_{[\mathcal{H}]}$

HSC state

multiple queries:

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\rho_{g\mathcal{H}g^{-1}})^{\otimes m}$$

Probability: $p_{[\mathcal{H}]}$

These states have been symmetricized!

Single HSC State

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \rho_{g\mathcal{H}g^{-1}} = \frac{1}{|\mathcal{G}|^2} \sum_{g \in \mathcal{G}} \sum_{h \in \mathcal{H}} D_R(ghg^{-1})$$

$$\rho_{[\mathcal{H}]} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} D_R(g) \rho_{\mathcal{H}} D_R(g^{-1})$$

$$\rho_{\mathcal{H}} \cong \bigoplus_{\mu} \left(I_{d_\mu} \otimes \frac{1}{|\mathcal{G}|} \sum_{h \in \mathcal{H}} D_\mu(h) \right)$$

Single HSC state is completely diagonalized (like Abelian!)

$$\rho_{[\mathcal{H}]} \cong \bigoplus_{\mu} c_\mu([\mathcal{H}]) \left(I_{d_\mu} \otimes I_{d_\mu} \right) \qquad c_\mu([\mathcal{H}]) = \frac{1}{d_\mu |\mathcal{G}|} \sum_{h \in \mathcal{H}} \chi_\mu(h)$$

Dihedral HSCP

$$\rho_{[\mathcal{H}]} \cong \bigoplus_{\mu} c_\mu([\mathcal{H}]) \left(I_{d_\mu} \otimes I_{d_\mu} \right) \qquad c_\mu([\mathcal{H}]) = \frac{1}{d_\mu |\mathcal{G}|} \sum_{h \in \mathcal{H}} \chi_\mu(h)$$

Dihedral group $\mathcal{D}_N$

2 one dimensional irreps

$$\chi_T(r^t s^k) = 1 \qquad \chi_A(r^t s^k) = (-1)^t$$

$\frac{N-1}{2}$ two dimensional irreps $\quad h \in \{1, 2, \ldots, \frac{N-1}{2}\}$

$$\chi_h(r^t s^k) = 2\delta_{t,0} \cos\left(\frac{2\pi h k}{N}\right)$$

	T	A	h
$c_\mu([\{e\}])$	$\frac{1}{2N}$	$\frac{1}{2N}$	$\frac{1}{2N}$
$c_\mu([\{e, r\}])$	$\frac{1}{N}$	0	$\frac{1}{2N}$

Optimal Measurement?



Pretty good measurement:

$$G_j = p_j T^{-1/2} \rho_j T^{-1/2} \qquad T = \sum_i p_i \rho_i$$

For Dihedral group HSCP, PGP is

$$G_{[\{e\}]} = [\frac{p}{2-p}] \oplus [1] \bigoplus_{h=1}^{\frac{N-1}{2}} (pI_2 \otimes I_2)$$

$$G_{[\{e,r\}]} = [\frac{2-2p}{2-p}] \oplus [0] \bigoplus_{h=1}^{\frac{N-1}{2}} ((1-p)I_2 \otimes I_2)$$

But this is not the optimal measurement!

Multiple Query HSC States

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} (\rho_{g\mathcal{H}g^{-1}})^{\otimes m}$$

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} D_R(g)^{\otimes m} \rho_{\mathcal{H}}^{\otimes m} D_R(g^{-1})^{\otimes m}$$

Diagonal representation of the group

$$D_R(g)^{\otimes m} = \left[\bigoplus_{\mu} (I_{d_\mu} \otimes D_\mu(g)) \right]^{\otimes m}$$

$$= \bigoplus_{\mu_1, \ldots, \mu_m} I_{d_{\mu_1}} \otimes \cdots I_{d_{\mu m}} \otimes D_{\mu_1}(g) \otimes \cdots \otimes D_{\mu_m}(g)$$

Clebsch-Gordan Decomposition

$$D_{\mu_1}(g) \otimes \cdots \otimes D_{\mu_m}(g) = \bigoplus_{\mu} \left(I_{n_\mu(\mu_1,\ldots,\mu_m)} \otimes D_\mu(g) \right)$$

Averaging over this representation,

$$\rho_{[\mathcal{H}]}^{\otimes m} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} D_R(g)^{\otimes m} \rho_{\mathcal{H}}^{\otimes m} D_R(g^{-1})^{\otimes m}$$

implies

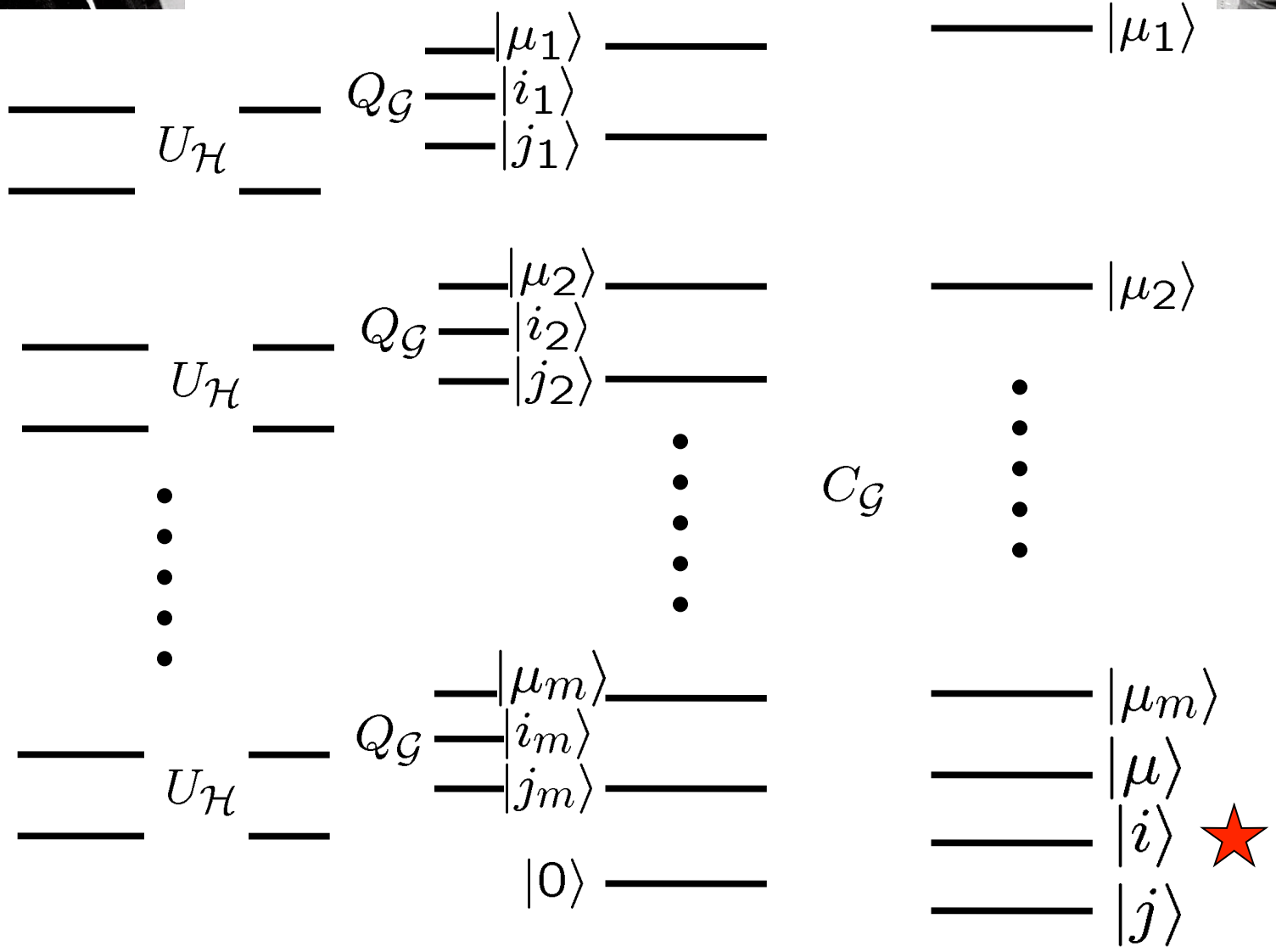
$$\rho_{[\mathcal{H}]}^{\otimes m} = \bigoplus_{\mu_1,\ldots,\mu_m} \left(I_D \otimes \bigoplus_{\mu} \left(R_{[\mathcal{H}]} \otimes I_{d_\mu} \right) \right)$$

$$D = d_{\mu_1} \cdots d_{\mu_m}$$



General Circuit





Quantum Algorithms for HSCP

1. Prepare *m* copies of the HS state.

2. Perform the QFT over $\mathcal{G}$ on each copy.
- efficient circuits for most groups of interest

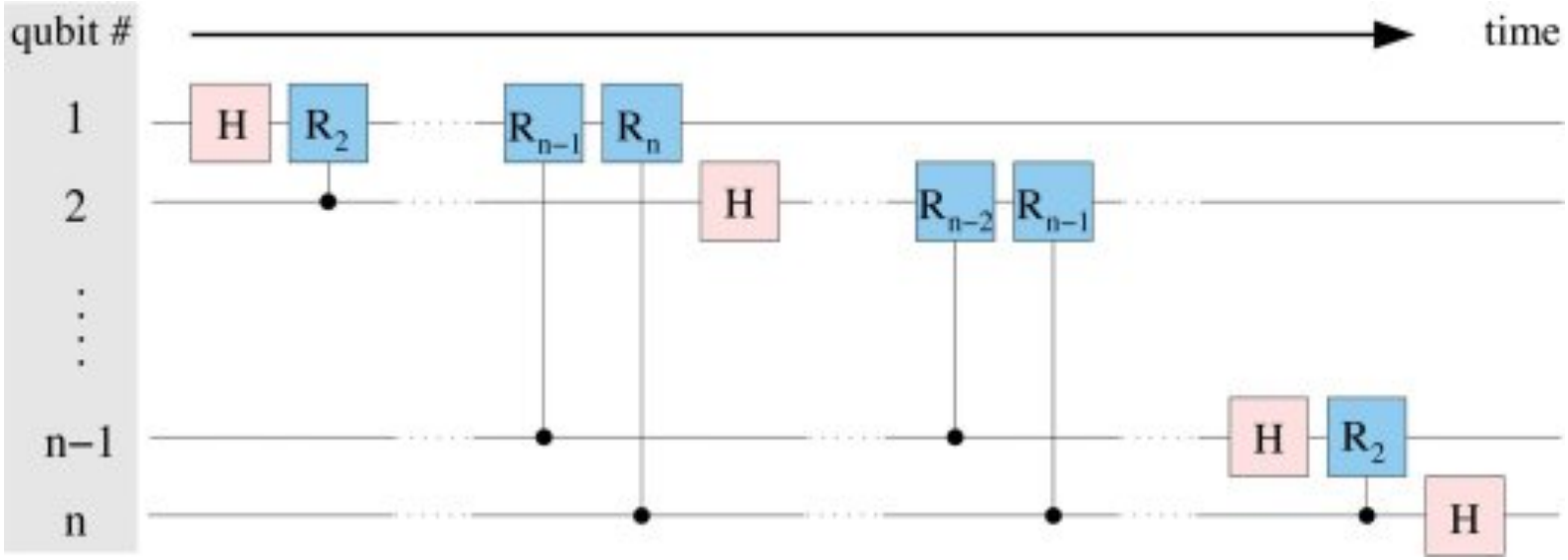
[Moore, Russell, Rockmore 03]

3. Perform CG transform on resulting states
- efficient circuits?

4. Measurement on multiplicity index
- efficient circuits?

The Second Role of Symmetry

Without efficient circuits for the quantum Fourier transform, all is for naught!



Second role of symmetries:

Subgroup adapted basis provide a recursive structure useful for constructing representation theoretic transforms

[Moore, Rockmore, Russel 03] Generic QFTs

[Bacon, Chuang, Harrow 04,06] Schur Transform

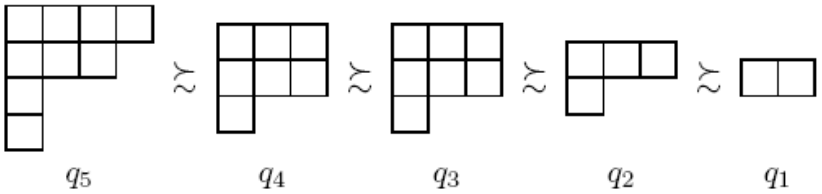
Clebsch-Gordan



[Bacon, Chuang, Harrow 04,06] Schur Transform

Clebsch-Gordan transform for U(d)

U(d) subgroup adapted basis (Gelfand-Zetlin basis)



Can these techniques be adapted for finite groups?

CG transform for U(d) is polynomial in d.

We want CG transform for regular representation of a finite group efficient in log of group size.

The To Do List!

Are the hidden subgroup conjugacy problem and the hidden subgroup problem equivalent?

For which groups are they equivalent?

Explicit understanding of the Clebsch-Gordan symmetry involved in the dihedral and symmetric group hidden subgroup conjugacy problem.

Construction of efficient circuits for finite group Clebsch-Gordan transforms.

Identification of optimal measurement for single copy hidden subgroup problem

QWI

group



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